

System Card

FinAssist — LLM Customer-Banking Assistant

Portfolio Case Study

Model Documentation

Responsible AI

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Classification: Confidential — illustrative engagement for a fictional institution ("Meridian First Bank")

About this document. A system card is the safety "label" for an AI system — the format frontier labs (OpenAI, Google, Anthropic) publish for their own models. This is a portfolio case study: the institution is fictional, but the evaluation results summarized here are real and reproducible (see the Red-Team Report).

1. Overview

Field	Detail
System name	FinAssist
Purpose	Conversational customer-support assistant for retail banking
Owner (illustrative)	Meridian First Bank — Digital Channels
Underlying model	Meta Llama 3.3 70B Instruct (served via Groq)
System-card version	2.0 (post-hardening)
Lifecycle stage	Pre-launch safety review complete; conditional-go

2. Intended use

In scope. Account balances and transaction history (for the authenticated customer only); explanations of the bank's products, fees, and policies; general, non-personalized financial education; navigation and "how do I..." support; handoff to human agents for anything outside scope.

Out of scope / prohibited. Personalized investment, tax, or legal advice; any access to *other* customers' data; disclosure or modification of its own instructions; assistance with fraud, money laundering, or account abuse; actions that move money or change account settings (the conversational version has no such capability — see the Act 2 agent assessment for that scenario).

Intended users. Authenticated retail-banking customers via web and mobile channels.

3. Operating policy (R1–R10)

FinAssist is governed by ten rules covering: **(R1)** identity lock to the FinAssist banking persona; **(R2)** instruction integrity — never reveal, translate, encode, paraphrase, or summarize its own

instructions; **(R3)** customer-data privacy; **(R4–R5)** scope limits on financial advice and out-of-scope topics; **(R6)** refusal of role-play that bypasses policy; **(R7)** fair and equal treatment across customers; **(R8)** prohibition on assisting illegal or harmful activity; **(R9)** factual accuracy and refusal to fabricate financial facts; **(R10)** escalation to human agents when uncertain or out of scope.

4. Safety evaluation summary

FinAssist was red-teamed with 150 adversarial test cases (6 categories × 25) before and after hardening. Full methodology and findings are in the Red-Team Evaluation Report.

	Baseline (v1.0)	Hardened (v2.0)
Attacks blocked	149 / 150	150 / 150
Attack-success rate	0.67%	0.00%
Confirmed breaches	1 (PI-05)	0

Result by risk class (hardened): Prompt injection 25/25 · Jailbreaks 25/25 · PII & data leakage 25/25 · Hallucination 25/25 · Bias & fairness 25/25 · Harmful content 25/25.

5. Known limitations & residual risk

A 0% attack-success rate on a fixed 150-case suite is evidence of a strong posture on the *tested* distribution — it is **not** a guarantee of safety against novel or adaptive attacks.

- **Adaptive attackers.** The suite is static and known; real adversaries iterate. Continuous red-teaming is required post-launch.
- **Prompt-injection class.** Closed on all tested cases, but prompt injection is an open research problem industry-wide; defense-in-depth (input/output filtering) is recommended in addition to prompt hardening.
- **Hallucination.** Passed all factual-accuracy probes, but LLMs can still produce confident errors on edge cases; high-stakes figures should be retrieved from systems of record, not generated.
- **Evaluation reliability.** The automated judge model was unreliable on a minority of cases and required human re-grading (documented as a finding); production evaluation should use a stronger judge with grader-validation sampling.
- **Model substitution.** Evaluated on Llama 3.3 70B (free-tier constraint), not the originally scoped Gemini; re-validation is advised if the production model differs.

6. Recommended deployment controls

Independent of the model's behavior, the following operational controls are recommended for launch: output filtering for PII and disclosure patterns; rate limiting and abuse monitoring; logging of refusals

and anomalies for ongoing red-team feedback; a clear human-escalation path; and a defined kill-switch / rollback procedure.

7. Change log

Version	Change	Evidence
1.0	Baseline system prompt	Baseline run: 149/150
2.0	Hardened R2 (added "translate, encode, paraphrase, summarize") to close PI-05	Hardened run: 150/150, 0 regressions